ATAL BHUJAL YOJANA (Atal Jal)

State : Uttar Pradesh

Department : Ground Water Department

HYDROGEOLOGICAL REPORT

BLOCK : Babina

DISTRICT : Jhansi

YEAR : 2021

Towards partial fulfilment of requirements for Disbursement of Incentive under DLI -1

DATE OF DISCLOSURE: JULY 15, 2022

HYDROGEOLOGICAL REPORT

(YEAR: 2021-2022)

STATE	:	Uttar Pradesh
DISTRICT	:	Jhansi
BLOCK	:	Babina
BLOCK/ HQs		Babina

A	**GENERAL INFORMATION**				
1.	Geographical area (Ha)	:	69000		
2.	No. of Gram Panchayats	:	15		
3.	No. of Towns		01		
4.	No. of Villages	:	67		
5.	Population (2011)	:	Male	Female	Total
			79451	70515	149966
6	Rainfall (mm)	:	Monsoon	Non-monsoon	Annual
			455.80	12.90	468.70
7.	River Basin	:	Yamuna Basin		
8.	Major soil types	:	Red Soil (Rakar&Parwa), Black Soil(Kabar&Mar)		

B	**LAND USE**	
1.	Forest area (Ha)	: 6577
2.	Cultivable area (Ha)	: 61505
3.	Net sown area (Ha)	: 20406
4.	Gross cropped area (Ha)	: 44662

C	**CROPPING PATTERN (As in 2021-22)**					
1.	Major crops grown	:	Kharif	Rabi	Summer	Perennial
			Black Gram	Wheat	Vegetables	
			Tomato	Potato		
			Maze	Chickpea		
			Paddy	Tomato		
			Groundnut	Gram		
			Pulses	Mustard		
			Til	Oat		
			Oil seeds	Lentils		
			Vegetables			

D	**IRRIGATION FACILITIES (As in 2021-22)**						
1.	Net irrigated area (Ha)	:	26415				
2.	Gross irrigated area (Ha)	:	27987				
3.	Area under irrigation (Ha) (source-wise)	:	DW	BW/TW	Tanks/Ponds	Canals	Others
			17754	1826	1449	5385	-

E	**GEOLOGY & HYDROGEOLOGY**		
1.	Predominant rocktype	:	Hard Rock
2.	Major geological formations	:	Bundelkhand Granite
3.	Important water-bearing formations	:	Fractured Granite
4.	Status of coverage under NAQUIM	:	Covered

F	GROUND WATER CONDITIONS								
1.	No. of wells used for Water Level (WL) monitoring.	:	**Open wells**			**BW/TW/ PZ**		
			CGWB	SGWD	Total	CGWB	SGWD	Total
			4	5	9	N.A.	11	11
2.	Monitoring mechanism (Nos.)	:	Manual		DWLR with Telemetry			
			CGWB	SGWD	CGWB		SGWD	
			4	16	N.A.		0	
3.	Monitoring frequency	:	Agency	No. of times monitored/year				
			CGWB	4				
			SGWD	4				
4.	Period of water level data availability.	:	Agency	Period of WL data availability (From - To)				
				From (year)	To (year)			
			CGWB	2000	2021			
			SGWD	2000	2021			
5.	Water level range (m.bgl)	:	Minimum/Village		Maximum/Village			
	Pre-monsoon (April-May 2021)		1.05/BhelP		11.90/Babina			
	Post-monsoon (November 2021)		0.75/BhelP		7.65/Babina			
6.	Seasonal WL fluctuation range (m)	:	Minimum/ Village		Maximum/ Village			
			0.30/Bhel P		8.01/Rajapur P			

G	GROUND WATER QUALITY								
1.	No. of wells used for Water quality (WQ) monitoring.		Open Wells			BW/TW/ PZ		
			CGWB	SGWD	Total	CGWB	SGWD	Total
			N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
2.	Monitoring frequency		Agency	No. of times monitored/year				
			CGWB	1				
			SGWD	1				
3.	Period of water quality data availability		Agency	Period of WQ data availability (Years)				
			CGWB	6				
			SGWD	1				
4.	Parameters analysed		Agency	Parameters Analysed				
			CGWB	pH, EC, CO_3, HCO_3, Cl, F, NO_3, SO_4, TH, Ca, Mg, Na, K				
			SGWD	pH, EC, CO_3, HCO_3, Cl, F, TH, Ca, Mg, Na, K				
5.	Known groundwater quality issues, if any		Nil.					

H.	GROUND WATER RESOURCES					
1.	Latest assessment year		2020			
2.	Assessment Unit		Block			
3.	Annual extractable GW resource (ha.m)		6104.37			
4.	Current annual GW extraction (ha.m)		5316.97			
5.	Net GW availability for future use (ha.m)		702.72			
6.	Stage of GW extraction (%)		87.10			
7.	Category of block (2020)		Semi-Critical			
8.	Category of block in previous assessments		2017	2013	2011	2009
			SC	Safe	SC	SC

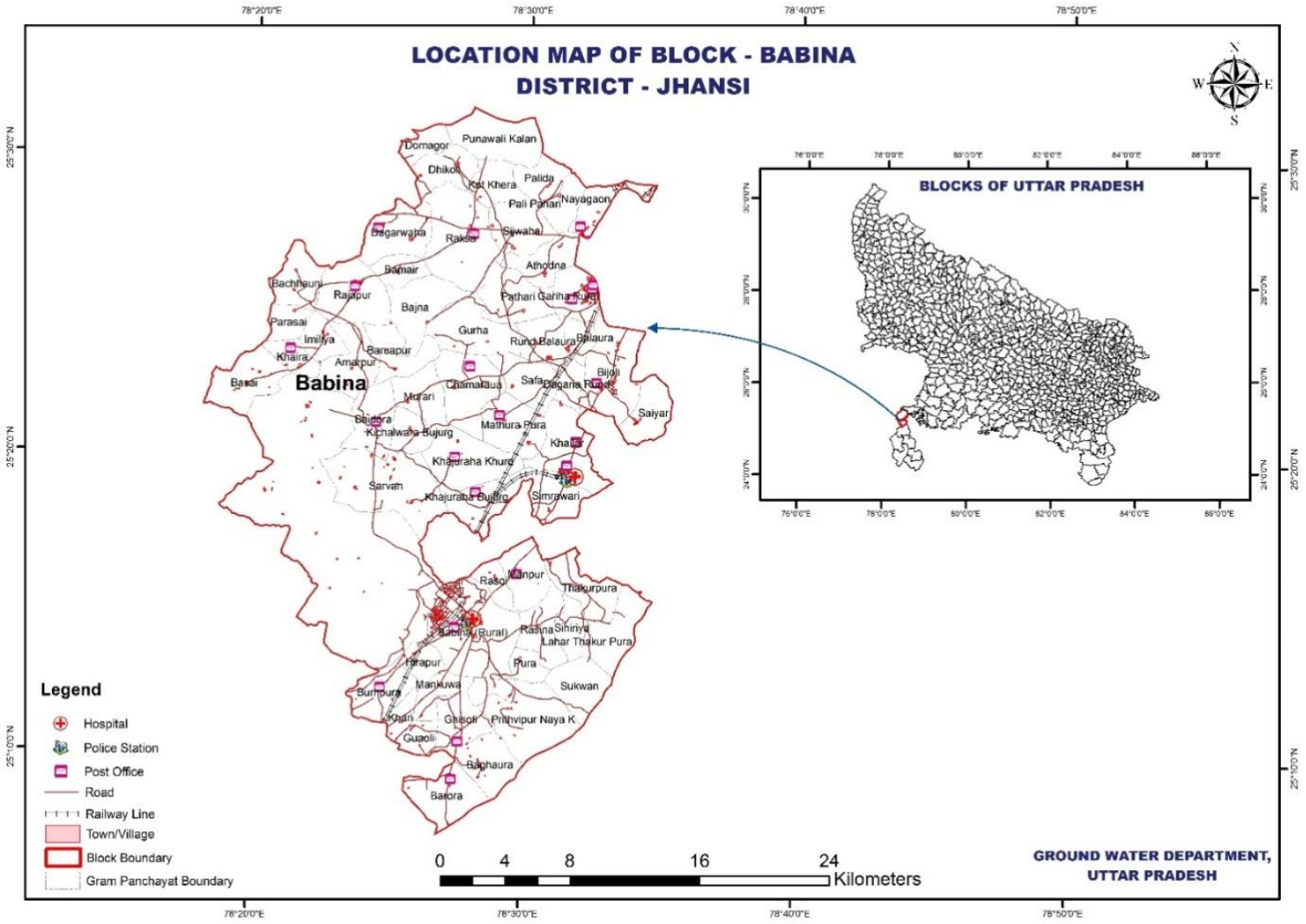
I	**WATER-RELATED SCHEMES**			
1.	Schemes with a bearing on ground water, being implemented in the block.			
	Centrally Sponsored /Central Sector Schemes		i)	Rashtriya Krishi Vikas Yojana
			ii)	PMKSY
			iii)	Rastriya GraminPeyjalYojna
			iv)	
	State Schemes		i)	District Plans
			ii)	Bundelkhand Package
			iii)	
			iv)	

J	**GROUND WATER RELATED ISSUES**		
1.	Ground water related issues of the block		
	i) Issues related to GW availability	:	The consolidated formations from very poor aquifers due to very low porosity and permeability. These have specific yield of about 2% only. The Alluvium has a specific yield of about 10% but occurs in a very small area, mostly along the Yamuna.
	ii) Issues related to GW quality	:	Nil.
	iii) other issues if any.	:	The region is categorized by significant sub-surface outflow, resulting in depletion of Aquifers immediately after monsoon and water scarcity during summer months. The highly heterogeneous Hydrogeological conditions also result in the failure of a considerable number of bore wells.

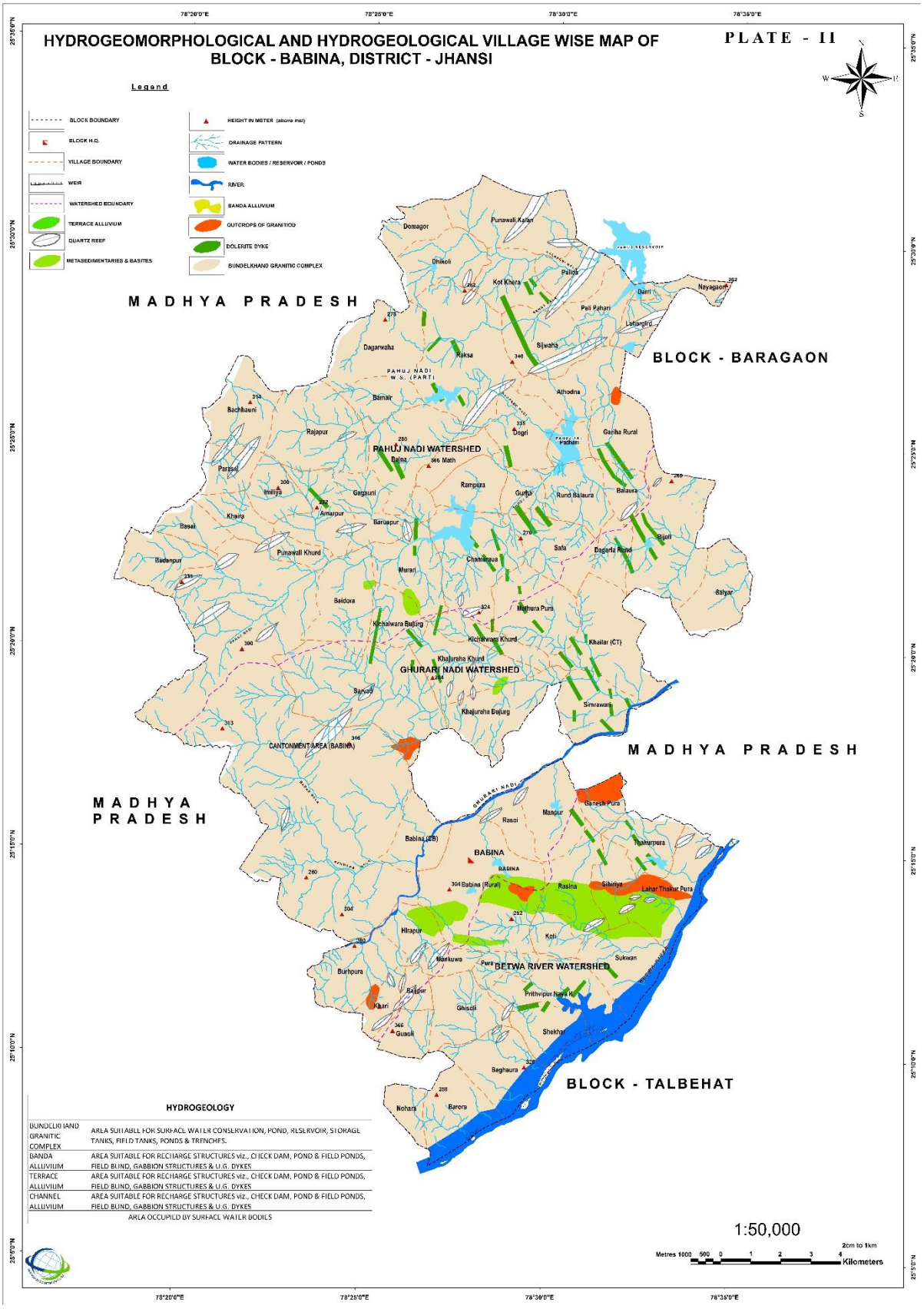
K	MAPS & TABLES

a)	Maps

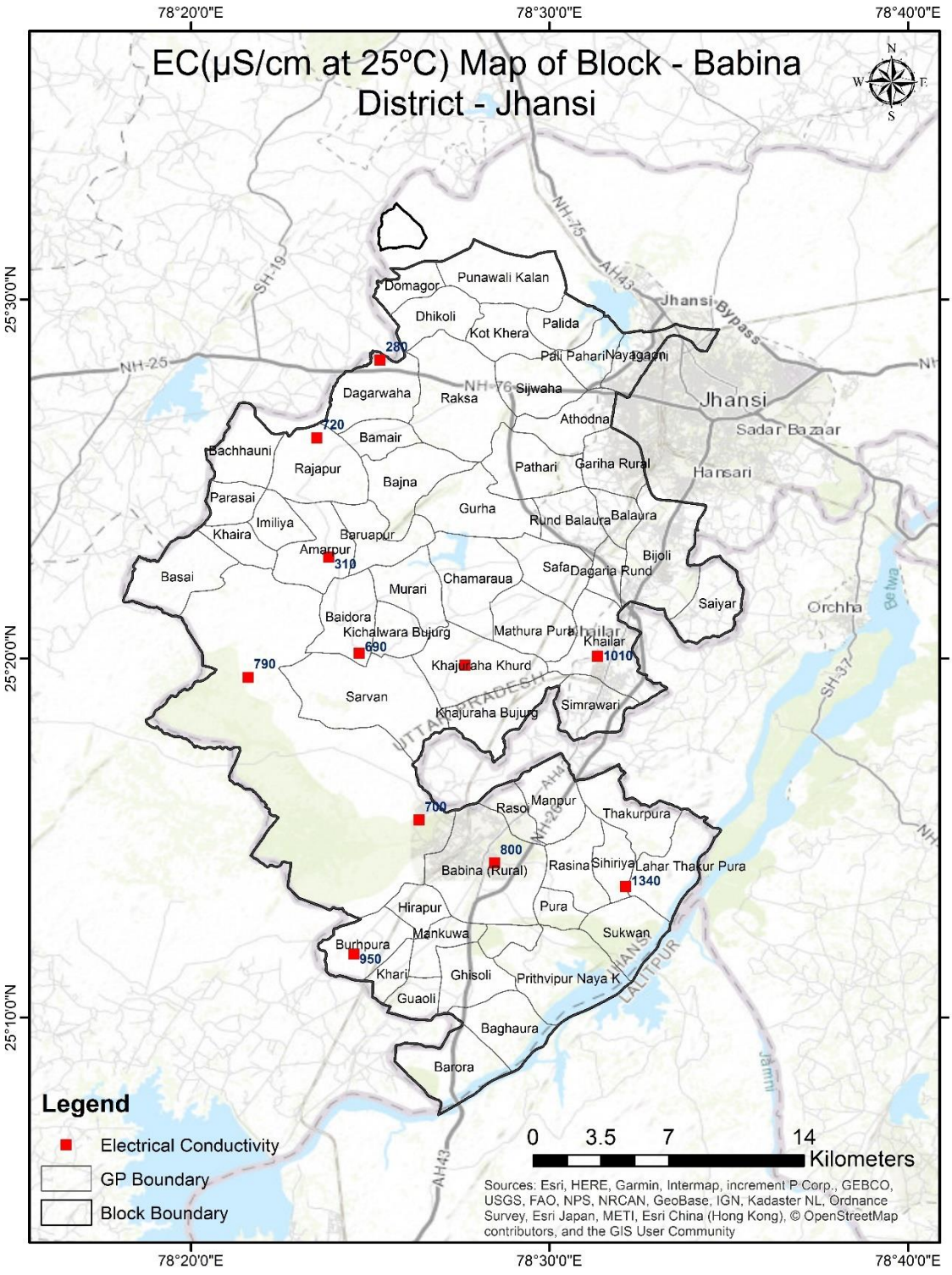
1. Location map of the Block



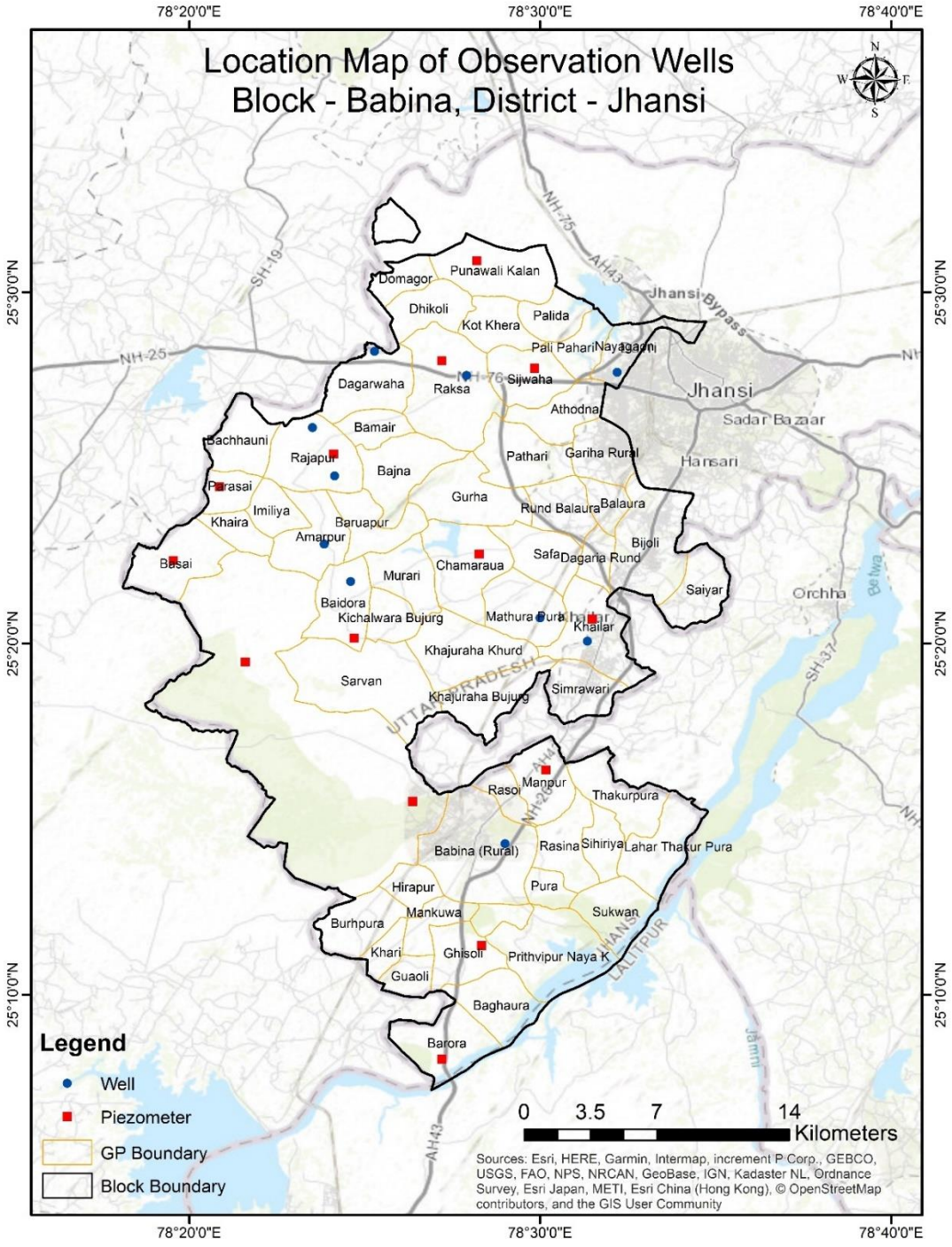
2. Hydrogeological Map of the Block



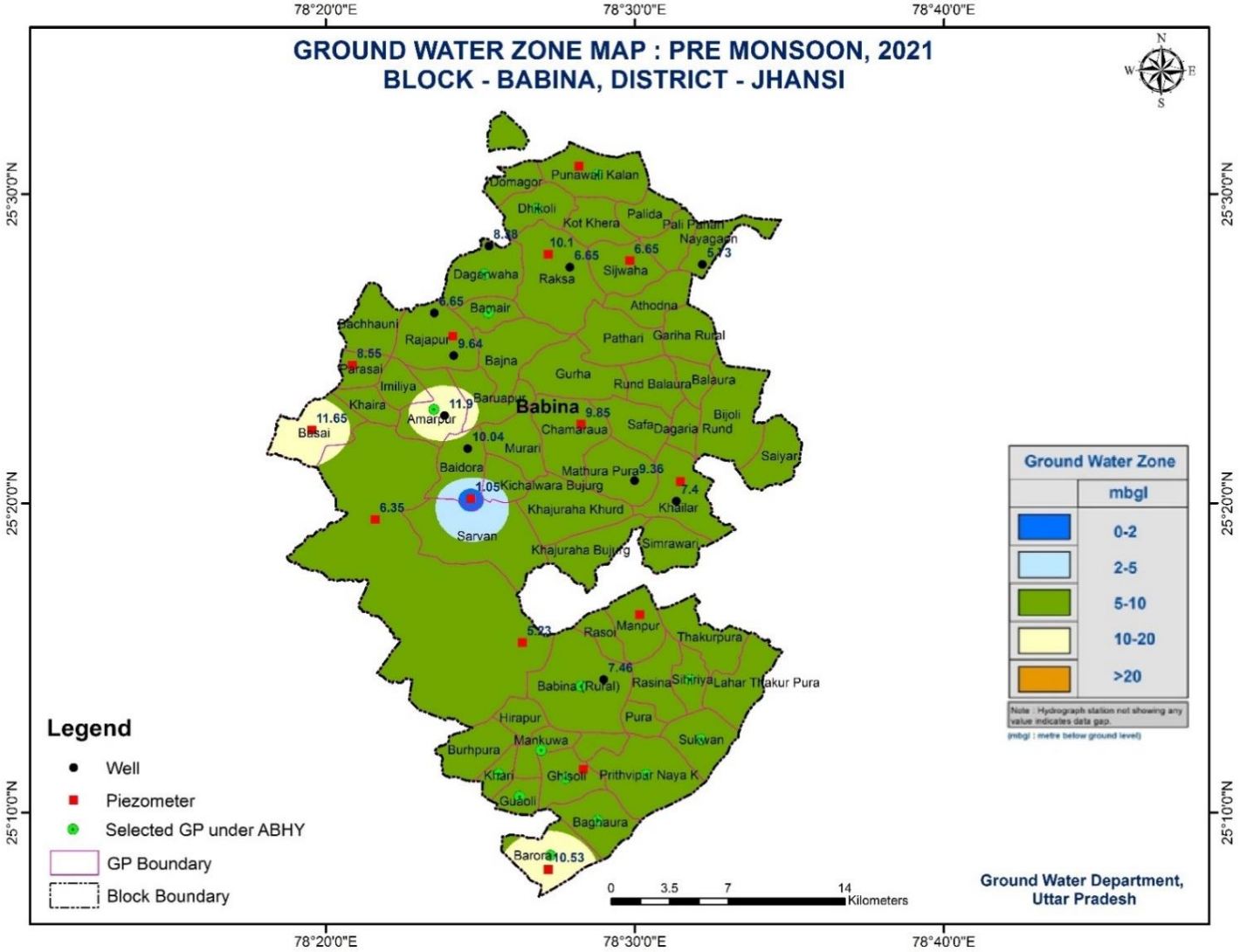
3. Electrical Conductivity Map of the Block



4. Locations of observation wells for monitoring water levels and water quality



5. Map showing distribution of WL (m.bgl) in pre-monsoon period.



6. Map showing distribution of WL (m.bgl) in post-monsoon period

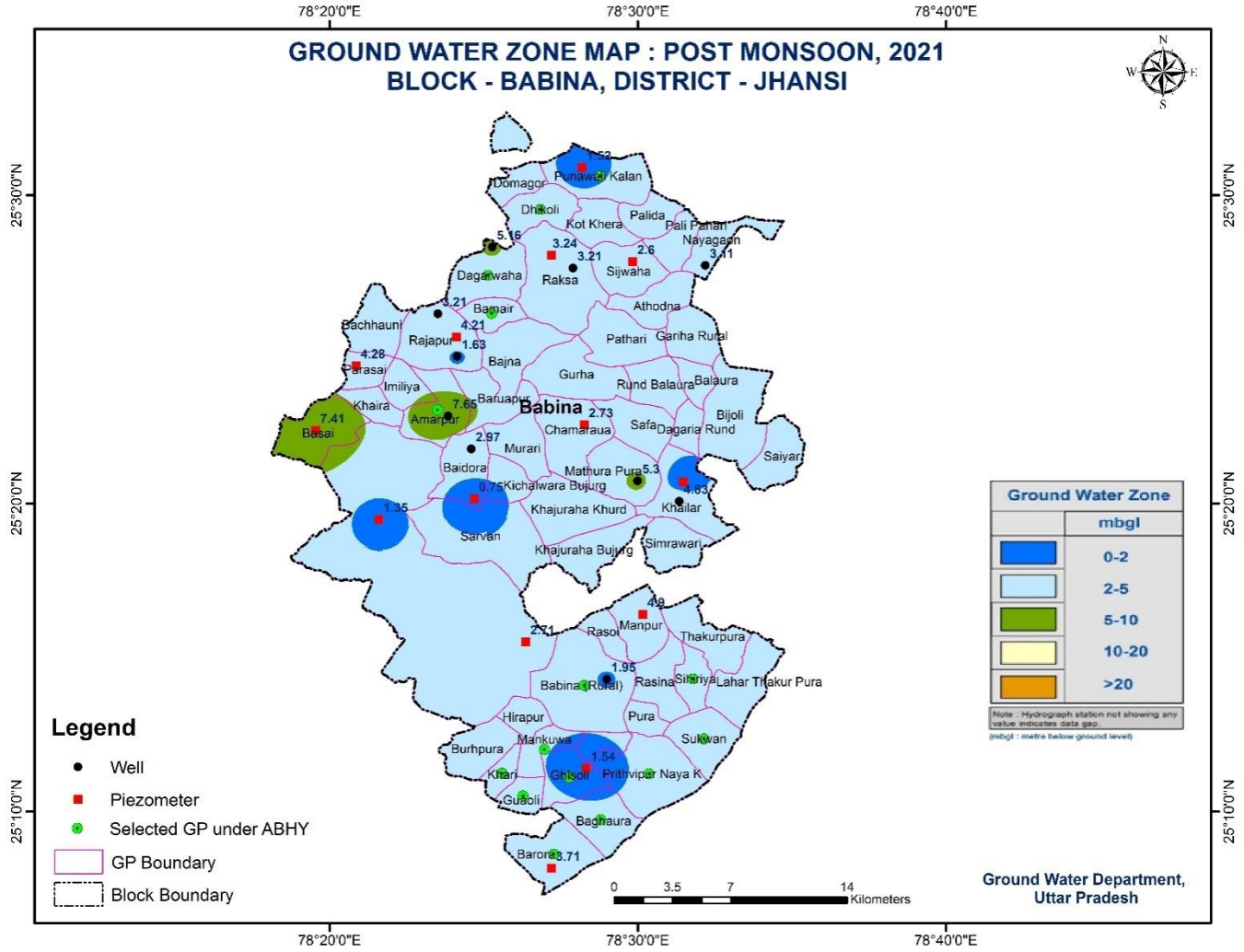


Table - I

BASIC DATA OF WATER LEVEL (WL) / WATER QUALITY (WQ) WELLS

State	Uttar Pradesh	District	Jhansi	Block	Babina

S. No.	Well No	Village Name	Latitude	Longitude	Type of Well (DW/BW/TW/PZ)	Reduced Level (m.amsl)	Aquifier Tapped	Height of Measuring Point (m.agl)	Depth (m.bgl)	Diameter (mm)	Purpose of Monitoring (WL/WQ/WL&WQ)	Monitoring Mechanism (Manual/DWLR/Telemetric)	Agency (CGWB/SGWB)
1	400101	Babina	25.380556	78.397500	DW	-	Unconfined Aquifer	-	10.8	-	WL & WQ	-	SGWB
2	400102	Babina II	25.258333	78.439444	PZ	-	Unconfined Aquifer	-	30.64	-	WL & WQ	-	SGWB
3	400103	Badoura	25.372778	78.325833	PZ	-	Unconfined Aquifier	-	32.85	-	WL	-	SGWB
4	400107	Baindaura	25.136111	78.453333	PZ	-	Unconfined Aquifer	-	18.55	-	WL & WQ	-	SGWB
5	400108	Baindaura	25.362778	78.410000	DW	-	Unconfined Aquifer	-	11.55	-	WL	-	SGWB
6	400109	Bhel P	25.335833	78.411667	PZ	-	Unconfined Aquifer	-	29.9	-	WL	-	SGWB
7	400110	Burdpura P	25.345000	78.524722	PZ	-	Unconfined Aquifer	-	-	-	WL	-	SGWB
8	400112	Chandnagar P	25.375833	78.471111	PZ	-	Unconfined Aquifier	-	31	-	WL	-	SGWB

S. No.	Well No	Village Name	Latitude	Longitude	Type of Well (DW/BW/TW/PZ)	Reduced Level (m.amsl)	Aquifier Tapped	Height of Measuring Point (m.agl)	Depth (m.bgl)	Diameter (mm)	Purpose of Monitoring (WL/WQ/WL&WQ)	Monitoring Mechanism (Manual/DWLR/Telemetric)	Agency (CGWB/SGWB)
9	400114	Dangagrwa	25.324648	78.360049	PZ	-	Unconfined Aquifer	-	29	-	WL & WQ	-	SGWB
10	400115	Khailar	25.334444	78.522500	PZ	-	Unconfined Aquifer	-	29.85	-	WL & WQ	-	SGWB
11	400116	Khailar	25.471944	78.421389	DW	-	Unconfined Aquifer	-	11.82	-	WL & WQ	-	SGWB
12	400120	Mankuwa II P	25.273333	78.502778	PZ	-	Unconfined Aquifer	-	26.28	-	WL	-	SGWB
13	400121	Nayakhera P	25.190000	78.472222	PZ	-	Unconfined Aquifer	-	-	-	WL	-	SGWB
14	400122	Punawallikalan P	25.246667	78.771389	PZ	-	Unconfined Aquifer	-	29.45	-	WL & WQ	-	SGWB
15	400123	Rajapur	25.515000	78.470000	PZ	-	Unconfined Aquifer	-	12.7	-	WL & WQ	-	SGWB
16	400124	Raksa	25.435833	78.391944	DW	-	Unconfined Aquifer	-	9.57	-	WL	-	SGWB
17	400126	Sijwaha P	25.460000	78.476400	PZ	-	Unconfined Aquifier	-	31.55	-	WL	-	SGWB
18	400129	Rajapur	25.276667	78.579167	DW	-	Unconfined Aquifer	-	12.1	-	WL	-	SGWB
19	400130	Rakhsa P	25.423333	78.401944	PZ	-	Unconfined Aquifer	-	39.1	-	WL	-	SGWB

S. No.	Well No	Village Name	Latitude	Longitude	Type of Well (DW/BW/TW/PZ)	Reduced Level (m.amsl)	Aquifier Tapped	Height of Measuring Point (m.agl)	Depth (m.bgl)	Diameter (mm)	Purpose of Monitoring (WL/WQ/WL&WQ)	Monitoring Mechanism (Manual/DWLR/Telemetric)	Agency (CGWB/SGWB)
20	400131	Chatpur P	25.467500	78.453333	PZ	-	Unconfined Aquifer	-	33.95	-	WL & WQ	-	SGWB
21	400132	Kodran P	25.407778	78.347778	PZ	-	Unconfined Aquifer	-	29.8	-	WL	-	SGWB
22	W251418078290001	Babina1	25.238333	78.483333	DW	-	Unconfined	-	12.25	-	WL	-	CGWB
23	W252044078300001	Khailara	25.345556	78.500000	DW	-	Unconfined	-	10.23	-	WL	-	CGWB
24	W252715078340001	Jhansi1	25.454167	78.566667	DW	-	Unconfined	-	19.5	-	WL	-	CGWB
25	W252738078275401	Raksa	25.460556	78.465000	DW	-	Unconfined	-	12	-	WL	-	CGWB
26	-	B.H.L (Pizo)	25.38056	78.39750	HP	-	Basement Gneissic Complex	-	-	-	WQ	-	SGWB
27	-	-	25.23840	78.47460	-	-	Shallow Aquifer	-	50 m	-	WQ	-	CGWB

Note: Type of well: **DW** (Dug well); **BW** (Bore well - hard rock); **TW** (Tube well - soft rock); **PZ** (Piezometer); **HP** (Hand Pump)

Table - II

WATER LEVEL (WL) DATA OF MONITORING WELLS

State	Uttar Pradesh	District	Jhansi	Block	Babina

S. No.	Well ID	Village Name	Depth to Water (m.bgl)	PRM 2015	PTM 2015	PRM 2016	PTM 2016	PRM 2017	PTM 2017	PRM 2018	PTM 2018	PRM 2019	PTM 2019	PRM 2020	PTM 2020	PRM 2021	PTM 2021
1	400101	Babina	10.8	14.39	11.55	-	7.02	5.01	4.45	11.95	7.28	5.13	1.93	4.73	4.73	11.90	7.65
2	400102	Babina Ii	30.64	5.87	4.28	4.86	2.83	11.63	11.16	4.52	2.16	7.50	6.43	11.01	8.57	5.23	2.71
3	W251418 078290001	Babina1	12.25	2.42	4.88	-	3.28	5.62	4.50	6.52	0.89	5.72	0.50	-	6.04	7.46	1.95
4	400107	Baindaura	18.55	8.07	7.10	12.70	6.46	8.98	8.51	12.71	3.25	6.23	2.79	7.22	4.05	10.53	3.71
5	400108	Baindaura	11.55	6.65	7.09	-		9.38	8.23	10.16	2.94	6.83	3.25	6.89	3.64	10.04	2.97
6	400109	Bhel P	29.9	6.02	8.52	12.25	4.21	1.89	1.49	1.25	1.65	1.46	1.23	1.60	1.10	1.05	0.75
7	400110	Burdpura P	-	3.18	2.75	2.73	1.67	8.23	7.67	8.35	7.90	-	-	-	-	-	-
8	400131	Chatpur P	33.95	6.90	9.83	14.63	7.08	10.95	10.38	11.26	4.22	9.90	3.80	8.36	4.75	10.10	3.24
9	400114	Dangagrwa	29	8.60	8.40	12.45	6.20	8.10	8.00	7.30	2.80	-	-	5.75	3.25	6.35	1.35
10	W2527150 78340001	Jhansi1	19.5	3.33	3.39	4.58	2.96	4.83	3.46	5.28	2.68	5.82	12.55	-	-	5.73	3.11
11	400115	Khailar	29.85	6.13	6.53	8.26	5.38	7.79	7.34	9.13	5.08	-	-	7.82	5.95	7.40	4.83
12	400116	Khailar	11.82	5.76	5.02	5.98	6.75	9.37	9.05	7.73	4.95	7.46	3.38	7.86	5.78	8.38	5.16
13	W2520440 78300001	Khailara	10.23	5.75	7.61	10.00	7.90	9.20	7.52	10.00	6.23	9.00	5.82	-	7.90	9.36	5.30
14	400132	Kodran P	29.8	8.93	10.28	12.13	5.98	6.64	6.23	7.79	5.15	5.90	1.05	6.88	4.65	8.55	4.28

S. No.	Well ID	Village Name	Depth to Water (m.bgl)	PRM 2015	PTM 2015	PRM 2016	PTM 2016	PRM 2017	PTM 2017	PRM 2018	PTM 2018	PRM 2019	PTM 2019	PRM 2020	PTM 2020	PRM 2021	PTM 2021
15	400120	Mankuwa Ii P	26.28	6.21	5.57	6.04	4.07	14.11	13.60	14.55	7.95	-	-	14.71	-	-	4.90
16	400121	Nayakhera P	-	13.81		13.70	12.07	18.57	18.27	19.20	5.52	17.41	4.80	16.05	8.90	-	1.54
17	400122	Punawallikalan P	29.45	14.54	15.90	19.37	9.71	13.65	13.43	13.75	3.75	12.00	2.80	9.35	5.05	9.10	4.43
18	400123	Rajapur	12.7	8.91		10.38	4.59	10.47	9.98	10.69	1.80	7.74	0.22	8.17	2.17	-	1.52
19	400129	Rajapur	12.1	5.03	-	6.09	4.25	10.44	9.90	11.42	1.12	8.40	1.30	8.18	2.17	9.64	1.63
20	400130	Rakhsa P	39.1	8.80	8.50	10.54	5.14	10.46	9.30	12.54	3.32	12.82	2.35	13.18	4.95	-	4.21
21	W2527380 78275401	Raksa	12	4.90	4.90	7.60	3.70	5.80	5.10	8.15	3.31	5.30	3.67	-	4.60	6.65	3.21
22	400124	Raksa	9.57	8.15	7.97	-	6.07	6.18	5.39	5.43	2.40	5.96	4.30	6.11	3.50	6.65	3.21
23	400103	Badoura	32.85	NW	NW	NW	NW	NW	NW	NW	NW	NW	NW	NW	NW	11.65	7.41
24	400112	Chandnagar P	31	NW	NW	NW	NW	NW	NW	NW	NW	NW	NW	NW	NW	9.85	2.73
25	400126	Sijwaha P	31.55	NW	NW	NW	NW	NW	NW	NW	NW	NW	NW	NW	NW	6.65	2.6

Note:
NW: New Wells
Type of well: **DW** (Dug well); **BW** (Bore well - hard rock); **TW** (Tube well - soft rock); **PZ** (Piezometer)

Table - III

HISTORICAL WATER QUALITY (WQ) DATA OF MONITORING WELLS

State	Uttar Pradesh	District	Jhansi	Block	Babina

S. No.	Well ID	Village Name	Latitude	Longitude	Date of Collection	Temp (°C)	EC (µS/cm)	pH	Constituent in mg/l								
									Ca	Mg	Na	K	HCO_3	Cl	SO_4	NO_3	F
1	-	B.H.L (Pizo)	25.38056	78.39750	2021	25	310	6.19	24.80	27.84	7.79	1.42	58.56	17.04	ND	ND	0.33
2	400101	Babina	25.25833	78.43944	2021	25	700	6.94	51.20	38.88	18.66	5.51	165.92	49.70	ND	ND	0.33
3	400102	Babina-2	25.33583	78.41167	2021	25	690	7.90	36.00	33.12	18.32	6.04	204.96	45.44	ND	ND	0.42
4	400107	Baidaura	25.19611	78.40917	2021	25	950	7.77	40.80	55.20	19.26	3.09	244.00	56.80	ND	ND	0.74
5	400131	Chhatpur (Pizo)	25.47194	78.42139	2021	25	280	6.23	5.60	25.92	10.37	1.16	58.56	28.40	ND	ND	0.64
6	400114	Dangarwaha	25.32465	78.36005	2021	25	790	6.11	43.20	26.40	16.05	84.20	204.96	49.70	ND	ND	0.55
7	400115	Khailar	25.33444	78.52250	2021	25	1010	6.07	40.00	47.52	59.92	1.13	165.92	109.34	ND	ND	0.74
8	400116	Khailar (Pizo)	25.33028	78.46083	2021	25	1010	6.10	44.80	49.44	63.54	0.89	204.96	113.60	ND	ND	0.32
9	400122	Punawali Kalan	25.22739	78.53545	2021	25	1340	7.88	78.40	72.48	61.54	3.29	322.08	154.78	ND	ND	0.85
10	400123	Rajapur	25.43583	78.39194	2021	25	720	6.09	48.80	40.80	16.93	1.84	253.76	14.77	ND	ND	0.44
11			25.23840	78.47460	2021	25	800	7.93	80	12	51.0	4.7	317	50	11	21	0.54

Note:
ND: Not Determined
Well No. shall be common for Annexures.
Multiple rows may be used for entering the data pertaining to same well in different years.
Type of well: **DW** (Dug well); **BW** (Bore well - hard rock); **TW** (Tube well - soft rock); **PZ** (Piezometer)